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RV COLLEGE OF ENGINEERING®

(An Autonomous Institution affiliated to VTU)

VI Semester B. E. Grade Improvement Examinations Nov-2020

Information Science and Engineering

SOFTWARE ENGINEERING AND TESTING

Time: 03 Hours ***Maximum Marks: 100***

Instructions to candidates:

1. Answer all questions from Part A. Part A questions should be answered in first three pages of the answer book only.
2. Answer FIVE full questions from Part B.In Part B question number 2, 7 and 8 are compulsory. Answer any one full question from 3 and 4 & one full question from 5 and 6.

PART A

1	1.1	Define Software Engineering.	01
	1.2	Define Software service.	01
	1.3	With a neat diagram, show the exploratory program development process.	02
	1.4	Mention the main factors contributing to the complexity of managing a software project.	02
	1.5	Mention the activities performed by project manager during project planning.	02
	1.6	Mention the import ways in which an experienced analyst gathers requirements.	02
	1.7	With a neat diagram, show the socio technical system stack.	02
	1.8	Mention the characteristics of a good software design.	02
	1.9	Differentiate between model-based versus modeless interface.	02
	1.10	Mention the reliability metrics of software products.	02
	1.11	Differentiate between verification and validation.	02

PART B

2	a	State the Lehman's laws.	08
	b	Explain the Classical Waterfall model.	08
3	a	Discuss the various software management complexities.	08
	b	Discuss the basic $COCOMO$ model.	08
		OR	
4	a	Explain the function point (FP) metric for project size estimation.	08
	b	Illustrate the focus areas of CMM levels and key process areas.	08
5	a	Briefly discuss the important ways in which an experienced analyst gathers requirements.	08
	b	List and explain the characteristics of a good SRS requirement.	08
		OR	

6	a	Discuss the organization of an *SRS* document as prescribed by *IEEE* 830 standard [*IEEE* 830].	10
	b	Describe context and interaction models in detail with examples.	06
7	a	Briefly discuss the different classes of cohersion.	08
	b	Explain the approaches to software design.	08
8	a	Discuss the reliability metrics of software products.	08
	b	Differentiate between black box testing and white box testing.	04
	c	With a neat diagram, explain the main activities of a testing process.	04